

Exam 2 Questions (7.A – 6.A)
CHEM 1310 | Spring 2026

QUESTION 1 (5 points)

A sample of an ideal gas at 21.2 °C and 0.992 atm in a rigid 4.00-L container is heated to 41.8 °C. What is the pressure of the gas in atmospheres (atm) at the end of this process? The ideal gas constant is 8.314 J/K·mol or 0.08206 L·atm/K·mol.

- A. ☒ **1.06 atm**
 - B. ☐ 0.932 atm
 - C. ☐ 1.96 atm
 - D. ☐ 0.503 atm
 - E. ☐ 1.12 atm
-

A sample of an ideal gas at 21.2 °C and 0.992 atm in a rigid 4.00-L container is cooled to 10.8 °C. What is the pressure of the gas in atmospheres (atm) at the end of this process? The ideal gas constant is 8.314 J/K·mol or 0.08206 L·atm/K·mol.

- A. ☒ **0.957 atm**
 - B. ☐ 0.505 atm
 - C. ☐ 1.95 atm
 - D. ☐ 1.13 atm
 - E. ☐ 1.06 atm
-

A sample of an ideal gas at 21.2 °C and 0.992 atm in a rigid 4.00-L container is cooled to 5.12 °C. What is the pressure of the gas in atmospheres (atm) at the end of this process? The ideal gas constant is 8.314 J/K·mol or 0.08206 L·atm/K·mol.

- A. ☒ **0.938 atm**
- B. ☐ 1.06 atm
- C. ☐ 4.11 atm
- D. ☐ 0.240 atm
- E. ☐ 1.13 atm

QUESTION 2 (6 points)

A gas cylinder with a volume of 50.0 L contains a mixture of He, Ar, and Kr gases at a total pressure of 30.0 atm. The partial pressures of the gases are as follows. Rank the gases from smallest (1) to greatest (3) number of moles in the mixture.

$$P(\text{He}) = 5.0 \text{ atm}$$

$$P(\text{Ar}) = 15.0 \text{ atm}$$

$$P(\text{Kr}) = 10.0 \text{ atm}$$

1 smallest number of moles	He
2	Kr
3 greatest number of moles	Ar

A gas cylinder with a volume of 50.0 L contains a mixture of He, Ar, and Kr gases at a total pressure of 30.0 atm. The partial pressures of the gases are as follows. Rank the gases from smallest (1) to greatest (3) number of moles in the mixture.

$$P(\text{He}) = 12.0 \text{ atm}$$

$$P(\text{Ar}) = 4.0 \text{ atm}$$

$$P(\text{Kr}) = 14.0 \text{ atm}$$

1 smallest number of moles	Ar
2	He
3 greatest number of moles	Kr

A gas cylinder with a volume of 50.0 L contains a mixture of He, Ar, and Kr gases at a total pressure of 30.0 atm. The partial pressures of the gases are as follows. Rank the gases from smallest (1) to greatest (3) number of moles in the mixture.

$$P(\text{He}) = 21.0 \text{ atm}$$

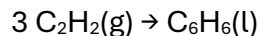
$$P(\text{Ar}) = 5.0 \text{ atm}$$

$$P(\text{Kr}) = 4.0 \text{ atm}$$

1 smallest number of moles	Kr
2	Ar
3 greatest number of moles	He

QUESTION 3 (5 points)

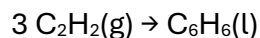
Gaseous acetylene (C_2H_2) can be converted to liquid benzene (C_6H_6) by certain catalysts according to the balanced chemical equation below.



Acetylene at a temperature of 250. K in a sealed vessel with a volume of 0.600 L and initial pressure of 1.62 atm was subjected to this reaction. The pressure when the reaction was terminated was 0.34 atm. Assuming no change in the volume of the gas, what amount in moles (mol) of benzene was produced in this process? Enter your response in moles (mol) to the nearest 0.001 mol.

0.012 mol

Gaseous acetylene (C_2H_2) can be converted to liquid benzene (C_6H_6) by certain catalysts according to the balanced chemical equation below.



Acetylene at a temperature of 250. K in a sealed vessel with a volume of 0.600 L and initial pressure of 3.62 atm was subjected to this reaction. The pressure when the reaction was terminated was 0.34 atm. Assuming no change in the volume of the gas, what amount in moles (mol) of benzene was produced in this process? Enter your response in moles (mol) to the nearest 0.001 mol.

0.032 mol

QUESTION 4 (5 points)

Which of the following statements is *true* regarding a 1.0 L sample of helium gas and a 1.0 L sample of argon gas at room temperature and atmospheric pressure? Consider both gases as ideal.

- A. ☐ The atoms in the helium sample will exhibit a narrower range of speeds than those in the argon sample.
 - B. ☐ The argon atoms exert a greater pressure on the walls of their container than the helium atoms.
 - C. ☐ **The average speed of atoms in the helium sample will be greater than the average speed of atoms in the argon sample.**
 - D. ☐ The volume of the helium sample is larger than the volume of the argon sample.
 - E. ☐ None of these statements is true.
-

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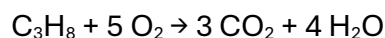
- A. ☐ **The atoms in the helium sample will exhibit a broader range of speeds than those in the argon sample.**
 - B. ☐ The argon atoms exert a greater pressure on the walls of their container than the helium atoms.
 - C. ☐ The average speed of atoms in the helium sample will be lower than the average speed of atoms in the argon sample.
 - D. ☐ The volume of the helium sample is larger than the volume of the argon sample.
 - E. ☐ None of these statements is true.
-

Which of the following statements is *true* regarding a 1.0 L sample of helium gas and a 1.0 L sample of argon gas at room temperature and atmospheric pressure? Consider both gases as ideal.

- A. ☐ The atoms in the helium sample will exhibit a narrower range of speeds than those in the argon sample.
- B. ☐ The argon atoms exert a greater pressure on the walls of their container than the helium atoms.
- C. ☐ The average speed of atoms in the helium sample will be lower than the average speed of atoms in the argon sample.
- D. ☐ The volume of the helium sample is larger than the volume of the argon sample.
- E. ☐ **None of these statements is true.**

QUESTION 5 (5 points)

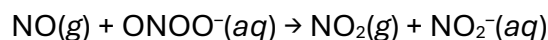
Formula Question. Combustion of propane (C_3H_8) proceeds according to the balanced chemical equation below. In a combustion reaction occurring at constant temperature, the rate of production of CO_2 is 0.036 M/min ([rate] M/min). What is the corresponding rate of consumption of C_3H_8 ? Enter your response in M/min to the nearest 0.001 M/min *with appropriate sign for a rate of consumption*.



-0.012 M/min = [rate] / -3

QUESTION 6 (6 points)

Use the dropdown menus below to specify the orders of NO and ONOO^- in the rate law for the given reaction. Then, indicate the units of the rate constant in the third dropdown.



Experiment	$[\text{NO}_2]_i$ (M)	$[\text{ONOO}^-]_i$ (M)	Initial Rate (M/min)
1	0.00125	0.00125	2.03×10^{-6}
2	0.00125	0.000625	1.02×10^{-6}
3	0.000625	0.00250	2.03×10^{-6}
4	0.000625	0.00375	3.05×10^{-6}

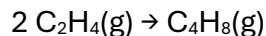
Order of NO: [☐ 0 ☒ 1 ☐ 2]

Order of ONOO^- : [☐ 0 ☒ 1 ☐ 2]

Units of the rate constant k : [☐ M/min ☐ 1/min ☒ 1/(M·min)]

QUESTION 7 (5 points)

The measurements below were made during the reaction of ethylene (C_2H_4) with itself, a second-order reaction with $k = 0.032 \text{ M}^{-1} \text{ s}^{-1}$.

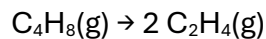


Determine the molarity of ethylene remaining at time $t = 300 \text{ s}$. Enter your response in moles per liter (M) to the nearest 0.001 M.

Time (s)	$[\text{C}_2\text{H}_4] \text{ (M)}$
0	0.100
100	0.076
200	0.061
300	?

0.051 M

The measurements below were made during the elimination of cyclobutane (C_4H_8), a first-order reaction with $k = 0.0032 \text{ s}^{-1}$.



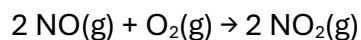
Determine the molarity of cyclobutane remaining at time $t = 300 \text{ s}$. Enter your response in moles per liter (M) to the nearest 0.001 M.

Time (s)	$[\text{C}_4\text{H}_8] \text{ (M)}$
0	0.100
100	0.073
200	0.053
300	?

0.038 M

QUESTION 8 (5 points)

Oxidation of NO occurs according to the balanced chemical equation and mechanism below. Fill in the blanks to describe this mechanism.



1. $2 \text{NO} + \text{O}_2 \rightarrow (\text{ONO})_2$
2. $(\text{ONO})_2 \rightarrow 2 \text{NO}_2$

The species $(\text{ONO})_2$ is a

[☐ reactant ☐ product ☐ catalyst ☐ **intermediate** ☐ transition state].

The rate law for the reaction is $\text{rate} = k[\text{NO}]^2[\text{O}_2]$. Based on this rate law, the slow step in the mechanism

[☐ is the first step. ☐ is the second step. ☐ **cannot be determined.**]

QUESTION 9 (5 points)

The reaction $\text{A(g)} + \text{B(g)} \rightarrow \text{C(g)}$ has a rate law: $\text{rate} = k[\text{A}]$. If the concentration of B is halved, what is the effect on the rate of the reaction?

- A. ☐ **The rate does not change.**
 - B. ☐ The rate doubles.
 - C. ☐ The rate increases by a factor of four.
 - D. ☐ The rate decreases by half.
 - E. ☐ The rate decreases by a factor of four.
-

The reaction $\text{A(g)} + \text{B(g)} \rightarrow \text{C(g)}$ has a rate law: $\text{rate} = k[\text{B}]^2$. If the concentration of B is halved, what is the effect on the rate of the reaction?

- A. ☐ **The rate decreases by a factor of four.**
 - B. ☐ The rate does not change.
 - C. ☐ The rate doubles.
 - D. ☐ The rate increases by a factor of four.
 - E. ☐ The rate decreases by half.
-

The reaction $A(g) + B(g) \rightarrow C(g)$ has a rate law: $\text{rate} = k[A]^2[B]$. If the concentration of B is halved, what is the effect on the rate of the reaction?

- A. ☐ The rate decreases by half.
- B. ☐ The rate decreases by a factor of four.
- C. ☐ The rate does not change.
- D. ☐ The rate doubles.
- E. ☐ The rate increases by a factor of four.

QUESTION 10 (5 points)

Consider two reactions **A** and **B** with equal activation energy, both occurring in a single elementary step. Reaction **A** has more productive orientations (i.e., orientations that lead to reaction upon collision) than reaction **B**. Which of the following statement(s) *must* be true?

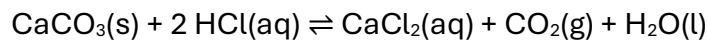
- [☐ Must be true ☐ **Could be false**] Reaction **A** is catalyzed and reaction **B** is not.
 - [☐ **Must be true** ☐ Could be false] The frequency factor of reaction **A** is greater than the frequency factor of reaction **B**.
 - [☐ Must be true ☐ **Could be false**] The overall order of reaction **A** is greater than the overall order of reaction **B**.
 - [☐ Must be true ☐ **Could be false**] The frequency factor of reaction **A** is equal to the frequency factor of reaction **B**.
-

Consider two reactions **A** and **B** with equal activation energy, both occurring in a single elementary step. Reaction **B** has more productive orientations (i.e., orientations that lead to reaction upon collision) than reaction **A**. Which of the following statement(s) *must* be true?

- [☐ **Must be true** ☐ Could be false] The frequency factor of reaction **A** is less than the frequency factor of reaction **B**.
- [☐ **Must be true** ☐ Could be false] At equal temperatures and concentrations, reaction **A** will proceed at a slower rate than reaction **B**.
- [☐ Must be true ☐ **Could be false**] The frequency factor of reaction **A** is greater than the frequency factor of reaction **B**.
- [☐ Must be true ☐ **Could be false**] Reaction **A** is catalyzed and reaction **B** is not.

QUESTION 11 (5 points)

Calcium carbonate is consumed in the presence of two equivalents of hydrochloric acid according to the balanced chemical equation below.



What is the correct equilibrium equation for this reaction?

- A. ☐ $K = \frac{[\text{CaCl}_2][\text{CO}_2]}{[\text{HCl}]^2}$
 $K = \frac{[\text{CaCl}_2][\text{CO}_2]}{[\text{HCl}]^2}$
- B. ☐ $K = \frac{[\text{CaCl}_2][\text{CO}_2][\text{H}_2\text{O}]}{[\text{CaCO}_3][\text{HCl}]^2}$
 $K = \frac{[\text{CaCl}_2][\text{CO}_2][\text{H}_2\text{O}]}{[\text{CaCO}_3][\text{HCl}]^2}$
- C. ☐ $K = \frac{[\text{HCl}]^2}{[\text{CaCl}_2][\text{CO}_2]}$
 $K = \frac{[\text{HCl}]^2}{[\text{CaCl}_2][\text{CO}_2]}$
- D. ☐ $K = \frac{[\text{CaCl}_2][\text{CO}_2]}{[\text{HCl}]}$
 $K = \frac{[\text{CaCl}_2][\text{CO}_2]}{[\text{HCl}]}$
- E. ☐ $K = \frac{[\text{CaCl}_2][\text{CO}_2][\text{HCl}]^2}{[\text{CaCl}_2][\text{CO}_2][\text{HCl}]^2}$
 $K = \frac{[\text{CaCl}_2][\text{CO}_2][\text{HCl}]^2}{[\text{CaCl}_2][\text{CO}_2][\text{HCl}]^2}$

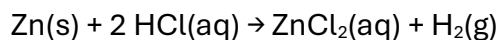
Solid potassium chlorate decomposes upon strong heating according to the following balanced chemical equation.



What is the correct equilibrium equation for this reaction?

- A. ☐ $K = [\text{O}_2]^3$
 $K = [\text{O}_2]^3$
- B. ☐ $K = \frac{[\text{KCl}][\text{O}_2]^3}{[\text{KClO}_3]}$
 $K = \frac{[\text{KCl}][\text{O}_2]^3}{[\text{KClO}_3]}$
- C. ☐ $K = \frac{[\text{KClO}_3]}{[\text{KCl}][\text{O}_2]^3}$
 $K = \frac{[\text{KClO}_3]}{[\text{KCl}][\text{O}_2]^3}$
- D. ☐ $K = \frac{1}{[\text{O}_2]^3}$
 $K = \frac{1}{[\text{O}_2]^3}$
- E. ☐ $K = \frac{[\text{KCl}][\text{O}_2]}{[\text{KClO}_3]}$
 $K = \frac{[\text{KCl}][\text{O}_2]}{[\text{KClO}_3]}$

Solid zinc reacts with aqueous hydrochloric acid to yield aqueous zinc chloride and hydrogen gas according to the following balanced chemical equation.



What is the correct equilibrium equation for this reaction?

- A. ☐ $K = \frac{[\text{H}_2][\text{ZnCl}_2]}{[\text{HCl}]^2}$
 $K = \frac{\text{[H}_2\text{] [ZnCl}_2\text{]}}{\text{[HCl]}^2}$
- B. ☐ $K = \frac{[\text{H}_2][\text{ZnCl}_2]}{[\text{HCl}]^2}$
 $K = \frac{\text{[HCl]}^2 \text{ [H}_2\text{] [ZnCl}_2\text{]}}{\text{[H}_2\text{] [ZnCl}_2\text{]}}$
- C. ☐ $K = \frac{[\text{H}_2][\text{ZnCl}_2]}{[\text{Zn}][\text{HCl}]^2}$
 $K = \frac{\text{[H}_2\text{] [ZnCl}_2\text{]}}{\text{[Zn] [HCl]}^2}$
- D. ☐ $K = \frac{[\text{Zn}][\text{HCl}]^2}{[\text{H}_2][\text{ZnCl}_2]}$
 $K = \frac{\text{[Zn] [HCl]}^2 \text{ [H}_2\text{] [ZnCl}_2\text{]}}{\text{[H}_2\text{] [ZnCl}_2\text{]}}$
- E. ☐ $K = \frac{[\text{ZnCl}_2]}{[\text{HCl}]^2}$
 $K = \frac{\text{[ZnCl}_2\text{]}}{\text{[HCl]}^2}$

QUESTION 12 (6 points)

Calcium carbonate decomposes according to the following balanced chemical equation. The equilibrium constant for this process is 1.8×10^7 .



Consider a reaction vessel containing 0.200 M CO_2 , 0.500 M CaCl_2 , and 0.0500 M HCl . To determine the equilibrium concentrations of reactants and products in this system, what expressions should appear in the Change row of an ICE table? x is the number of reaction occurrences between the initial state and equilibrium. Enter "0" if the system is already in equilibrium.

HCl: [☐ -2x ☐ -x ☐ 0 ☐ +x ☐ +2x]

CaCl₂: [☐ -2x ☐ -x ☐ 0 ☐ +x ☐ +2x]

CO₂: [☐ -2x ☐ -x ☐ 0 ☐ +x ☐ +2x]

Calcium carbonate decomposes according to the following balanced chemical equation. The equilibrium constant for this process is 1.8×10^7 .



Consider a reaction vessel containing 0.200 M CO_2 , 0.500 M CaCl_2 , and 1.0×10^{-5} M HCl . To determine the equilibrium concentrations of reactants and products in this system, what expressions should appear in the Change row of an ICE table? x is the number of reaction occurrences between the initial state and equilibrium. Enter "0" if the system is already in equilibrium.

HCl: [☐ $-2x$ ☐ $-x$ ☐ 0 ☐ $+x$ ☐ $+2x$]

CaCl₂: [☐ $-2x$ ☐ $-x$ ☐ 0 ☐ $+x$ ☐ $+2x$]

CO₂: [☐ $-2x$ ☐ $-x$ ☐ 0 ☐ $+x$ ☐ $+2x$]

Calcium carbonate decomposes according to the following balanced chemical equation. The equilibrium constant for this process is 1.8×10^7 .



Consider a reaction vessel containing 0.200 M CO_2 , 0.900 M CaCl_2 , and 1.0×10^{-4} M HCl . To determine the equilibrium concentrations of reactants and products in this system, what expressions should appear in the Change row of an ICE table? x is the number of reaction occurrences between the initial state and equilibrium. Enter "0" if the system is already in equilibrium.

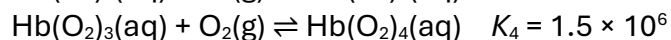
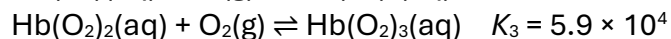
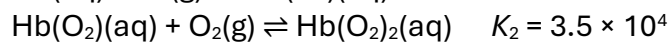
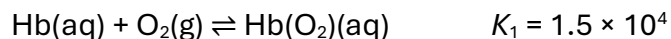
HCl: [☐ $-2x$ ☐ $-x$ ☐ 0 ☐ $+x$ ☐ $+2x$]

CaCl₂: [☐ $-2x$ ☐ $-x$ ☐ 0 ☐ $+x$ ☐ $+2x$]

CO₂: [☐ $-2x$ ☐ $-x$ ☐ 0 ☐ $+x$ ☐ $+2x$]

QUESTION 13 (5 points)

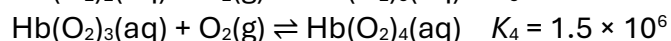
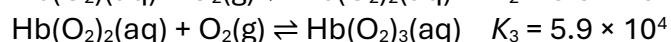
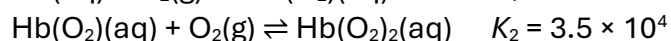
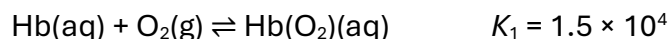
Hemoglobin (Hb) is associated with four oxygen-binding equilibria.



What is the equilibrium constant for the reaction $\text{Hb(O}_2\text{)}_4\text{(aq)} \rightleftharpoons \text{Hb(aq)} + 4 \text{O}_2(\text{g})$?

- A. ☐ 2.2×10^{-20}
 - B. ☐ 4.6×10^{19}
 - C. ☐ 3.5×10^8
 - D. ☐ 1.1×10^{-5}
 - E. ☐ 6.7×10^{-24}
-

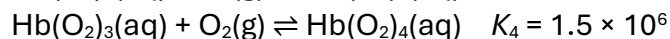
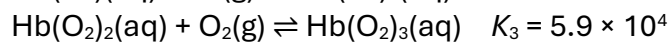
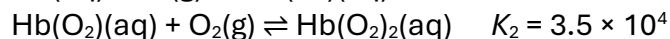
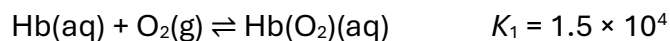
Hemoglobin (Hb) is associated with four oxygen-binding equilibria.



What is the equilibrium constant for the reaction $\text{Hb(O}_2\text{)}_2\text{(aq)} \rightleftharpoons \text{Hb(aq)} + 2 \text{O}_2(\text{g})$?

- A. ☐ 1.9×10^{-9}
 - B. ☐ 5.3×10^8
 - C. ☐ 1.5×10^4
 - D. ☐ 3.5×10^4
 - E. ☐ 3.6×10^{-14}
-

Hemoglobin (Hb) is associated with four oxygen-binding equilibria.

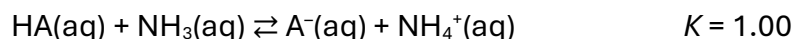


What is the equilibrium constant for the reaction $\text{Hb(O}_2\text{)(aq)} \rightleftharpoons \text{Hb(aq)} + \text{O}_2(\text{g})$?

- A. ☐ 6.7×10^{-5}
- B. ☐ 8.6×10^8
- C. ☐ 1.5×10^4
- D. ☐ 2.3×10^{-4}
- E. ☐ 3.5×10^4

QUESTION 14 (5 points)

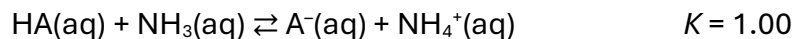
The reaction of acetic acid (HA) with ammonia (NH₃) in aqueous solution is very close to perfectly reversible.



In a solution initially containing 0.500 M HA and 0.200 M NH₃, what is the equilibrium concentration of acetate ion (A⁻)? Enter your response in moles per liter (mol/L, M) to the nearest 0.001 M.

0.143 M

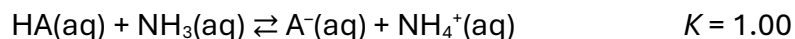
The reaction of acetic acid (HA) with ammonia (NH₃) in aqueous solution is very close to perfectly reversible.



In a solution initially containing 0.400 M HA and 0.400 M NH₃, what is the equilibrium concentration of acetate ion (A⁻)? Enter your response in moles per liter (mol/L, M) to the nearest 0.001 M.

0.200 M

The reaction of acetic acid (HA) with ammonia (NH₃) in aqueous solution is very close to perfectly reversible.

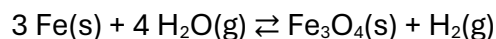


In a solution initially containing 0.500 M HA and 0.500 M NH₃, what is the equilibrium concentration of acetate ion (A⁻)? Enter your response in moles per liter (mol/L, M) to the nearest 0.001 M.

0.250 M

QUESTION 15 (6 points)

Consider a system in equilibrium with respect to the reaction below. Indicate whether each of the following interventions will cause the generation of more reactants (←), more products (→), or neither (⇌). The reaction is exothermic with $\Delta H^\circ = -151 \text{ kJ/mol}$.

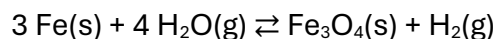


[☐ ← ☐ → ☐ ⇌] Decreasing the reaction temperature

[☐ ← ☐ → ☐ ⇌] Removing H₂(g) from the reaction vessel

[☐ ← ☐ → ☐ ⇌] Decreasing the volume of the reaction vessel

Consider a system in equilibrium with respect to the reaction below. Indicate whether each of the following interventions will cause the generation of more reactants (←), more products (→), or neither (⇌). The reaction is exothermic with $\Delta H^\circ = -151 \text{ kJ/mol}$.

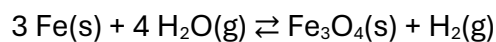


[☐ ← ☐ → ☐ ⇌] Adding Fe(s) to the reaction vessel

[☐ ← ☐ → ☐ ⇌] Increasing the reaction temperature

[☐ ← ☐ → ☐ ⇌] Removing H₂O(g) from the reaction vessel

Consider a system in equilibrium with respect to the reaction below. Indicate whether each of the following interventions will cause the generation of more reactants ($\leftarrow$), more products ($\rightarrow$), or neither ($\rightleftharpoons$). The reaction is exothermic with $\Delta H^\circ = -151 \text{ kJ/mol}$.



[☐ $\leftarrow$ ☐ $\rightarrow$ ☐ $\rightleftharpoons$] Increasing the reaction temperature

[☐ $\leftarrow$ ☐ $\rightarrow$ ☐ $\rightleftharpoons$] Adding $\text{Fe}_3\text{O}_4\text{(s)}$ to the reaction vessel

[☐ $\leftarrow$ ☐ $\rightarrow$ ☐ $\rightleftharpoons$] Increasing the volume of the reaction vessel

QUESTION 16 (5 points)

What is the correct chemical formula for the conjugate base of HCrO_4^- ?

- A. ☐ CrO_4^{2-}
 - B. ☐ H_2CrO_4
 - C. ☐ HCrO_4^{2-}
 - D. ☐ HCrO_4
 - E. ☐ HCrO_3^-
-

What is the correct chemical formula for the conjugate acid of HCrO_4^- ?

- A. ☐ H_2CrO_4
 - B. ☐ CrO_4^{2-}
 - C. ☐ HCrO_4^{2-}
 - D. ☐ HCrO_4
 - E. ☐ HCrO_3^-
-

What is the correct chemical formula for the conjugate acid of MnO_4^- ?

- A. ☐ HMnO_4
- B. ☐ MnO_4^{2-}
- C. ☐ MnO_4
- D. ☐ MnO_3^-
- E. ☐ HMnO_4^+

QUESTION 17 (5 points)

What is the pH of a solution made by dissolving $[N] \times 10^{-3}$ mol $\text{Sr}(\text{OH})_2$ in enough water to make 3.00 L of solution? Enter your response to the nearest 0.01.

$$14 + \log(2 * N / 3.00) - 3$$

$$N = 1.00 \text{ to } 5.00$$

QUESTION 18 (6 points)

Indicate whether each of the following is a state variable or path variable.

[☐ state variable ☒ **path variable**] Work

[☐ **state variable** ☐ path variable] Energy

[☐ **state variable** ☐ path variable] Volume

Indicate whether each of the following is a state variable or path variable.

[☐ **state variable** ☐ path variable] Temperature

[☐ **state variable** ☐ path variable] Pressure

[☐ state variable ☒ **path variable**] Heat

QUESTION 19 (5 points)

A gas is contained in a cylinder capped with a movable piston. The external pressure on the piston is constant. The gas is heated, causing it to expand and push the piston outward. Which of the following statements is *true* concerning the work done by the gas during this process?

- A. ☒ **The work done by the gas is positive because the system expands against the external pressure. NOTE: by the typical convention of chemists, this (correct) reasoning would lead to negative work. Credit was given for both this answer choice and choice B.**
 - B. ☐ The work done by the gas is negative because heat is added to the system.
 - C. ☐ The work done on the gas is positive because the piston moves outward.
 - D. ☐ No work is done because the external pressure remains constant.
 - E. ☐ The work done by the gas is zero because only the temperature changes.
-

A gas is contained in a cylinder capped with a movable piston. The external pressure on the piston is constant. The gas is cooled, causing it to contract and the piston to pull inward. Which of the following statements is *true* concerning the work done by the gas during this process?

- A. ☐ The work done by the gas is negative because the system contracts against the external pressure. **NOTE:** by the typical convention of chemists, this (correct) reasoning would lead to positive work. Credit was given for both this answer choice and choice B.
 - B. ☐ The work done by the gas is positive because heat is removed from the system.
 - C. ☐ The work done on the gas is negative because the piston moves inward.
 - D. ☐ No work is done because the external pressure remains constant.
 - E. ☐ The work done by the gas is zero because only the temperature changes.
-

A gas is contained in a cylinder capped with a movable piston. The external pressure on the piston is constant. The gas is cooled and more gas flows into the cylinder, causing its temperature to drop while its volume remains constant. Which of the following statements is *true* concerning the work done by the gas during this process?

- A. ☐ The work done by the gas is negative because the temperature decreases.
- B. ☐ The work done by the gas is positive because heat is removed from the system.
- C. ☐ The work done on the gas is negative because more gas is added to the system.
- D. ☐ No work is done because the external pressure remains constant.
- E. ☐ No work is done because the volume of the gas remains constant. **NOTE:** with no change in volume of the gas, no work is done on or by the gas. Only this choice is correct.